Drive systems & mechanisms

ENGINEERING WITH MR KELLY

Higher · Mechanisms & structures · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name the four types of motion and give examples
- ☐ calculate velocity ratio, output speed and torque in gear trains
- ☐ describe the effect of an idler gear
- ☐ design a compound gear train to give a required speed
- ☐ describe ways to reduce friction and wear
- ☐ calculate torque, power and rotational speed in drive systems

I DO — Worked example

Read how a full-mark answer is built

An idler gear B is placed between gears A and C.

1 State the effect on the output speed if the number of teeth on idler B is reduced. (1 mark)

Full-mark answer each ✓ is one mark

There is no effect on the output speed; an idler only changes the direction of rotation ✓.

Why answers lose marks

"It reverses the direction of gear C."

0 marks: the question asks about speed.

WE DO — Guided practice

Use the starters and word bank to help you

Moving parts in a drive system wear quickly.

2 Describe one way to reduce wear. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

friction · grease · harder · self-lubricating · reduce · lubricate · bearings · materials

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A 20-tooth driver turns a 60-tooth gear. The VR is...
☐ 1:3 ☐ 3:1 ☐ 20:1 ☐ 60:1

b) Reducing the teeth on an idler gear makes the output...
☐ Faster ☐ Slower ☐ No different ☐ Reverse

c) A force of 50 N at a radius of 0.2 m gives a torque of...
☐ 10 N m ☐ 250 N m ☐ 0.004 N m ☐ 50.2 N m

d) A pneumatic piston moving in and out shows...
☐ Rotary motion ☐ Oscillating motion ☐ Reciprocating motion ☐ Linear only

4 Key terms: write the letter of the matching description next to each term.

1. **Reciprocating motion**	____	**A** Makes driver and driven turn the same way; doesn't change the VR.
2. **Oscillating motion**	____	**B** Speed of input ÷ speed of output = driven teeth ÷ driver teeth.
3. **Velocity ratio**	____	**C** Input speed × input teeth = output speed × output teeth.
4. **Gear speed method**	____	**D** Swinging backwards and forwards in an arc.
5. **Idler gear**	____	**E** Backwards and forwards in a straight line.
6. **Compound gear train**	____	**F** Two gears on one shaft; total VR = VR1 × VR2.

A motor produces 2.4 N m of torque at 1200 revs min⁻¹.

5 Calculate the power output. (2 marks)

A winch drum of diameter 0.2 m lifts a 50 kg load.

6 Calculate the torque on the drum. (2 marks)

A motorised winch is driven by a motor at 1200 revs min⁻¹. The drum must turn at 25 revs min⁻¹.

7 Calculate the velocity ratio required. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?